

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0327

Electric 100 kV
March 21, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0327. The Interconnection Customer proposes a 126.8 MW natural gas combined cycle facility interconnecting at the Electric 100 kV substation in the Illinois area. The request was received on April 23, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 229 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,160,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Electric to Shell Chemical 100.0 kV Line 1
Contingency: Shell Chemical to Electric 100.0 kV line
Loading: 124.6% (Rating: 141 MVA)
DFAX: 87.5%
Pre-project loading: 46.0%
2. Facility: Electric to Shell Chemical 100.0 kV
Contingency: Shell Chemical to Electric 100.0 kV Line
Loading: 124.6% (Rating: 141 MVA)
DFAX: 87.5%
Pre-project loading: 46.0%
3. Facility: Revere Dr to Corona Ave 100.0 kV Line 1
Contingency: Revere Dr to Corona Ave 100.0 kV Circuit 2
Loading: 100.1% (Rating: 154 MVA)
DFAX: 10.2%
Pre-project loading: 91.8%
4. Facility: Revere Dr to Corona Ave 100.0 kV Line
Contingency: Revere Dr Substation to Corona Ave 100.0 kV Cir. 1
Loading: 100.1% (Rating: 154 MVA)
DFAX: 10.2%
Pre-project loading: 91.8%
5. Facility: Revere Dr to Corona Ave 100.0 kV Line 3
Contingency: Corona Ave to Revere Dr 100.0 kV #1
Loading: 100.1% (Rating: 154 MVA)
DFAX: 10.2%
Pre-project loading: 91.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Electric to Shell Chemical 100.0 kV Line 1	Rebuild line with higher-capacity conductor	\$609,000	100% (\$609,000)
Electric to Shell Chemical 100.0 kV	Rebuild line with higher-capacity conductor	\$1,138,000	100% (\$1,138,000)
Revere Dr to Corona Ave 100.0 kV Line 1	Reconductor line	\$1,201,000	100% (\$1,201,000)
Revere Dr to Corona Ave 100.0 kV #2	Reconductor line	\$835,000	100% (\$835,000)
Revere Dr to Corona Ave 100.0 kV Line 3	Reconductor line	\$377,000	100% (\$377,000)
	Total allocated to project		\$4,160,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.